



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

Instruction Division

**First Semester 2026-27
Course Handout Part II**

Date: 01-08-2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No.: ME U423
Course Title: Micro Fluidics And Its Applications
Instructor In-charge: Prof. A R Harikrishnan
Tutorial Instructors: -NA

Course Description:

This elective course introduces the fundamental principles and emerging applications of microfluidics. This is one of the dominating areas in prime movers. The course emphasizes both theoretical foundations and practical insights relevant to microscale fluid behavior, fabrication methods and devices.

Scope and Objective of the Course:

The scope of this course spans across fundamental science, device engineering, and emerging real-world applications at the micro- and nano-scale. This course provides a comprehensive understanding of microscale fluid dynamics, and control principles central to microfluidic systems. The course objectives are to understand the basics of microfluidics and the physics governing microfluidic flows and interfaces, to explore various microfabrication techniques suitable for fluidic devices and to analyze and understand the microfluidic systems applications in various engineering domains.

Prerequisite: Basic fluid Mechanics understanding

Text Books:

(T1). Nguyen, N. T., Werely, S. T., “Fundamentals and applications of Microfluidics”, Artech house Inc., 2002.

Reference Books:

- R1. Bruus, H., “Theoretical Microfluidics”, Oxford University Press Inc., 2008.
- R2. Tabeling, P., “Introduction to microfluidics”, Oxford University Press Inc., 2005..
- R3. Kirby, B.J., “Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices”, Cambridge University Press, 2010.
- R4. Colin, S., “Microfluidics”, John Wiley & Sons, 2009



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Course Plan:

Module	Lecture	Learning Objective	Topic to be discussed/ Learning outcomes	Ref. to text
Introduction to Microfluidics	1-2	Introduction to microfluidics	What is microfluidics, importance and applications, Challenges, Commercial activities	T1, Ch1, R1 Ch1
	3-4	Scaling in microfluidics	Physics of miniaturization, Scaling laws	T1, Ch1, R1 Ch1, R2 Ch1
Micro-scale fluid mechanics and pressure driven microflows	5-6	Governing equations and Constitutive relations	Intermolecular forces, States of matter, Continuum assumption, Governing equations, Constitutive relations.	T1, Ch2, R1 Ch2
	7-13	Micro scale flows and pressure driven microflows	Low Re flows, Slip theory, Entrance effects, Stokes drag on a sphere, Exact solutions, Couette flow, Poiseuille flow,	T1, Ch2, R1 Ch3, R2 Ch2, class notes
	14-18	Droplet Microfluidics	Bubble and droplet generation; annular and slug-annular flow regimes	R1 Ch13, R2 Ch2, class notes
Capillary flows	19-22	Capillarity and Interfacial Phenomenon	Surface tension and interfacial energy, Young-Laplace equation, Contact angle, Capillary length and capillary rise, Interfacial boundary conditions, Marangoni effect	R1 Ch7, class notes
Electrokinetics	23-27	Electrokinetics and Electro osmotic flows	Electrohydrodynamics fundamentals. Electro-osmosis, Debye layer, Thin EDL limit, Ideal electroosmotic flow (EOF), Ideal EOF with back pressure, EOF of power-law fluids.	R1 Ch8, ch9, R2 Ch4, class notes
	28-32	Electrophoresis and Dielectrophoresis	Introduction to the concept of electrophoresis and dielectrophoresis, Direct electro-wetting, Electro-wetting on dielectric.	T1 Ch2, R1 Ch10, R2 Ch4, class notes



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Microfabrication techniques	33-37	Material and methods for micro fabrication	Materials, Clean room, Silicon crystallography, Miller indices. Oxidation, photolithography- mask, spin coating, exposure and development, Etching, Bulk and Surface micromachining, Wafer bonding. Polymer microfabrication, PMMA/COC/PDMS substrates.	T1 Ch3, R2 Ch6
Microfluidics components and applications	38-40	Microfluidics components and applications of microfluidics, Lab visits	Introduction to some microfluidic components such as micropumps, droplet generators, mixers and understand application of microfluidics in some practical applications, lab visits	T2 Ch3, R2 Ch7

Evaluation Scheme

Components	Duration	Weightage (%)	Date & Time	Remarks
Mid-semester exam	90 min.	30	10/10/2025	Closed book
Minor projects*	-	25	Continuous	Open book
Assignment	-	10	Will be announced in class	Open book
Comprehensive exam	3 hrs.	35	13/12/2025	Partially closed and partially open book

* The students will be encouraged to design and conduct some experiments in the lab as part of the course assignments/projects to have experiential learning.

Notices: All notices related to this course will be updated on the Nalanda. The students should also check their emails regularly.

Make-up Policy: It will be as per the institute rules. Only genuine cases will be considered. No makeup will be provided for the quizzes under any circumstances.

Instructor-in-charge

Prof. A R Harikrishnan

ME F423

